



University of Central Punjab
Faculty of Information Technology and

Week No. 2

Lab Manual: Information

Security **Lab Title:** Classical

Ciphers (Code) **Lab**

Objectives

By the end of this lab, students will be able to:

1. Perform cryptanalysis on classical substitution ciphers.
2. Use frequency analysis to identify cipher patterns.
3. Demonstrate the process of recovering plaintext without knowing the key (Brute force).

Learning Outcomes (CLOs & PLOs Mapping)

CLO	CLO Statement	Bloom's Level	PLO
1	Apply Information Security techniques into practice.	P3 (Apply)	PLO-5

Lab Overview

This lab introduces Cryptanalysis, the art and science of breaking ciphers without knowing the key.



Experiment 1: Understanding Frequency Distribution

Theory:

Every language has a characteristic letter frequency.

For example, in English, the letters E, T, and A appear most

frequently. Letter Frequency (%)

E 12.7

T 9.1

A 8.2

O 7.5

I 7.0

N 6.7

Task:

1. Take a sample text (plaintext).
2. Count the occurrence of each alphabet letter.
3. Plot or print the frequency table.



```
#include <iostream>
using namespace std;

int main()
{
    string text;
    int count[26] = {0};

    cout << "Enter text: ";
    getline(cin, text);

    for(int i = 0; i < text.length(); i++)
    {
        char ch = text[i];

        if(ch >= 'a' && ch <= 'z')
            ch = ch - 32;

        if(ch >= 'A' && ch <= 'Z')
            count[ch - 'A']++;
    }

    for(int i = 0; i < 26; i++)
    {
        if(count[i] > 0)
            cout << char('A' + i) << " = " << count[i] << endl;
    }

    return 0;
}
```

Task:

- Enter your Full Name as input.
- Perform frequency analysis to find the most repeated letter.
- **Expected Output:** A table showing each letter's frequency percentage.



Experiment 2: Frequency Analysis on Caesar Cipher

Theory:

Since Caesar Cipher is a simple shift cipher, the **frequency distribution** of the ciphertext matches that of plaintext — only shifted.

By identifying the most frequent letter in ciphertext and comparing it to expected frequencies (like 'E'), the **shift key** can be deduced.

Task:

- Ciphertext: "SOJZKXS KDGS OY ZUU KGYE"
- Perform frequency analysis to find the most probable key and recover plaintext.